

1	(a) force proportional to product of the two masses and inversely proportional to the square of their separation <i>either</i> reference to point masses <i>or</i> separation $\gg$ 'size' of masses	M1	A1	[2]
	(b) gravitational force provides the centripetal force $GMm/R^2 = mR\omega^2$ where m is the mass of the planet $GM = R^3\omega^2$	B1	M1	A1
		A0		[3]
	(c) $\omega = 2\pi / T$ <i>either</i> $M_{\text{star}} / M_{\text{Sun}} = (R_{\text{star}} / R_{\text{Sun}})^3 \times (T_{\text{Sun}} / T_{\text{star}})^2$ $M_{\text{star}} = 4^3 \times (1/2)^2 \times 2.0 \times 10^{30}$ $= 3.2 \times 10^{31} \text{ kg}$ <i>or</i> $M_{\text{star}} = (2\pi)^2 R_{\text{star}}^3 / GT^2$ $= \{(2\pi)^2 \times (6.0 \times 10^{11})^3\} / \{6.67 \times 10^{-11} \times (2 \times 365 \times 24 \times 3600)^2\}$ $= 3.2 \times 10^{31} \text{ kg}$	C1	C1	A1
				[3]
		(C1)		
		(C1)		
		(A1)		